

Karim Yasser

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Summary

Software Engineer with over 2 years of experience building scalable applications across mobile, desktop, and web platforms. Proficient in **Python, C++**, **Kotlin**, **Dart**, and **TypeScript** with a strong foundation in **data structures**, **algorithms**, and **system design**. Proven expertise in clean architecture, agile methodologies, database optimization, and delivering production-ready solutions from conceptualization to deployment.

Education

CAIRO UNIVERSITY, FACULTY OF ENGINEERING Sep 2022 - Jun 2027
Bachelor of Engineering in **Computer Engineering**, GPA: Excellent

Experience

Software Developer - iSUPPLY Jul 2025 - Dec 2025

- Won first place in a 30+ team hackathon, earning a full-time position offer.
- Collaborated with back-end engineers and QA testers in an agile environment, analyzing the Flutter desktop app's architecture and state management to guide feature development.
- Implemented offline functionality and optimized complex MySQL database queries with direct backend integration for a production system handling over 1000 products.

Android Developer Intern - Banque Misr Jul 2025 - Sep 2025

- Developed native Android applications using Kotlin, Android Jetpack components, and the Room Database for local data persistence.
- Implemented MVVM architecture patterns to create scalable and maintainable mobile solutions with enhanced user experience and optimized performance.

Mobile Development Intern (Flutter & ML) - Informatique Jul 2025 - Aug 2025

- Integrated Machine Learning models into Flutter applications to deliver responsive, intelligent mobile solutions.
- Developed cross-platform features by bridging Dart with ML frameworks (e.g., TFLite/ML Kit) for on-device data processing.

Session Lead - Udacity (Digital Egypt Cubs Initiative) Nov 2025 - Mar 2026

- Led weekly sessions for student cohorts, delivering core computer fundamentals and system design principles.
- Supported learners through guided code reviews, practical examples, and structured technical discussions.

Projects

Aladdin: Nasira's Revenge - C++ Game Engine Mar 2026 - May 2026 | [GitHub](#)

Developed a high-performance 3D engine using **C++17** and **OpenGL 3.3** featuring a data-driven **Entity Component System (ECS)** with deferred lifecycle management. Engineered a **command-based renderer** with **16k shadow mapping** and integrated **miniaudio** and **FFmpeg** for disk-streamed media.

Rusty Sherlook - High-Performance Search Engine Feb 2026 - Mar 2026 | [GitHub](#)

Engineered a high-concurrency extension to the Sherlook search engine in **Rust** utilizing **Tokio**. Integrated **Qdrant** for semantic vector search and **Neo4j** for advanced graph-based PageRank processing.

iSupply POS App - First Place Hackathon Winner Jun 2025 | [GitHub](#)

Developed a cross-platform **Flutter POS system** for desktop and mobile, featuring inventory search, cart management, and invoice tracking. Integrated **Hive** for local storage and **Supabase** backend with offline synchronization.

AI API Evaluation Framework Mar 2026 - Apr 2026 | [GitHub](#)

Engineered a high-performance **FastAPI** orchestrator with **SSE streaming** for real-time monitoring of AI agents. Developed an **Agent Trace Viewer** for multi-turn loops, enabling granular credit assignment across reasoning and tool-calling steps.

Autonomous Wall-Following Robot

Apr 2026 - May 2026 | [GitHub](#)

Developed professional-grade **bare-metal C** firmware for ATmega328P utilizing **Timer1** and **EXTI/PCINT** interrupts for non-blocking sensing. Designed a **Cascaded PID Controller** nesting a wall-distance loop within an IMU heading loop.

Yapper - Cross-Platform Mobile App

Oct 2025 - Dec 2025 | [GitHub](#)

Built a full-stack social media application utilizing **React Native** and **Expo**. Implemented real-time messaging via **WebSockets**, multi-language **RTL** support, and automated deployments with a comprehensive **CI/CD** pipeline.

Technical Skills

- **Programming Languages:** Python, C++ , C, Kotlin, Dart, Java, TypeScript, JavaScript, SQL
- **Frameworks & Libraries:** Flutter Bloc, Riverpod, Dagger/Hilt, Room, Retrofit, FastAPI, Scikit-learn, Pandas, Node.js
- **Backend & Database:** Supabase, Firebase, MySQL, PostgreSQL, SQLite, Offline-first Architecture, API Integration
- **Development Tools:** Git, GitHub, Docker, Linux, VS Code, CI/CD, Agile/Scrum
- **Software Engineering:** Clean Architecture, MVVM, SOLID Principles, Dependency Injection, State Management, Object-Oriented Programming (OOP), Design Patterns

Core Competencies

- **Problem Solving:** Analytical thinking, Problem decomposition, Algorithm design, Debugging, Root cause identification
- **Leadership:** Technical mentoring, Code review, Cross-functional collaboration, Agile teamwork
- **Communication:** Technical documentation, Knowledge transfer, Remote collaboration
- **Continuous Learning:** Technology adoption, Industry best practices, Rapid skill acquisition